

Reading Bilirubin Correctly.

Bilirubin is the oldest test in hepatology, and it is the one we read most carelessly. A laboratory report gives a total bilirubin, and often a second number called direct or conjugated bilirubin. Many clinicians look at the total, decide whether the child is yellow enough to worry about, and move on. That habit misses the most important information on the report.

A bilirubin result only makes sense when you know three things. First, which fraction is raised: the unconjugated part, which has not yet been processed by the liver, or the conjugated part, which has. Second, which laboratory method produced the number, because two methods that sound the same give very different results. Third, which step in the handling of bilirubin has failed, because each step has its own diseases.

In this chapter we will first follow a molecule of bilirubin from its birth in a red cell to its exit into the gut. Then we will look at the laboratory tests and at how much a raised value is worth. After that we will compare what five guidelines call abnormal in a newborn, and see why the old twenty percent rule should be abandoned. Finally we will study the inherited disorders: Gilbert syndrome and Crigler-Najjar syndrome on the unconjugated side, and Dubin-Johnson and Rotor syndromes on the conjugated side.

A note on units. In this chapter, bilirubin values are given in milligrams per decilitre unless I say otherwise. Many laboratories outside the United States report micromoles per litre. To convert milligrams per decilitre to micromoles per litre, multiply by about seventeen. So one milligram per decilitre is about seventeen micromoles per litre, and five milligrams per decilitre is about eighty-five.

Part one. Five steps, five kinds of disease.

Bilirubin is a waste product. It comes from haem, the iron-containing ring found in haemoglobin and in a few other proteins. The body cannot use it, and must get rid of it. It does this in five steps, one after another, like stations on a production line. Because the steps happen in order, the step that fails leaves its own signature on the blood test. If you know the five steps, you can usually name the problem from the numbers alone.

The first step is production. When old red cells are removed, mainly in the spleen, haem is opened by an enzyme called haem oxygenase. This produces a green pigment, biliverdin, together with iron and carbon monoxide. A second enzyme, biliverdin reductase, turns biliverdin into bilirubin. This bilirubin is unconjugated. It does not dissolve in water. When production goes up, the unconjugated bilirubin goes up. The usual causes are haemolysis, ineffective production of red cells in the bone

marrow, and the breakdown of a large bruise or haematoma, such as a cephalhaematoma in a newborn. With haemolysis, other tests move at the same time: the reticulocyte count and LDH go up, and haptoglobin goes down.

The second step is transport. Because unconjugated bilirubin does not dissolve in water, it travels in the blood tightly bound to albumin. Albumin has one high-affinity binding site for bilirubin, and each gram of albumin can carry roughly nine milligrams of bilirubin. The small fraction that is not bound is called free bilirubin. Free bilirubin is the dangerous part, because it can cross into the brain. This step explains a trap. If albumin is low, or if a drug pushes bilirubin off albumin, the total bilirubin on the report may not change at all, while the free bilirubin rises. Two drugs that displace bilirubin are well known in neonatal care: sulfonamides and ceftriaxone. So in a jaundiced newborn you should always look at the albumin next to the bilirubin.

The third step is uptake into the liver cell. On the side of the liver cell that faces the blood, called the sinusoidal side, two transport proteins take bilirubin in. They are called OATP one B one and OATP one B three. They also take back conjugated bilirubin that has leaked into the blood, as we will see in a moment. When they are both missing, the result is Rotor syndrome, a rare and harmless cause of raised conjugated bilirubin with normal liver enzymes.

The fourth step is conjugation. Inside the liver cell, the enzyme UGT one A one attaches one or two molecules of glucuronic acid to bilirubin. The result is called bilirubin monoglucuronide and diglucuronide, or together, conjugated bilirubin. Conjugated bilirubin dissolves in water. UGT one A one is the only enzyme that matters for this job. When it is weak or absent, unconjugated bilirubin rises and the conjugated fraction stays low. The causes are Gilbert syndrome, Crigler-Najjar syndrome types one and two, and the normal immaturity of the enzyme in newborns.

The fifth step is excretion. On the side of the liver cell that faces the bile canaliculus, a pump called MRP two pushes conjugated bilirubin out into bile. This pump uses energy, and it is the slowest step in the whole chain. The liver cell also has a second pump, MRP three, on the blood side. It can push some conjugated bilirubin back into the blood, where liver cells further along the sinusoid take it up again through the OATP transporters. This is why, when the OATP transporters are missing in Rotor syndrome, conjugated bilirubin accumulates in the blood. When excretion fails, conjugated bilirubin rises. The causes are Dubin-Johnson syndrome, where MRP two itself is missing, and, much more commonly, every form of cholestasis, whether it comes from inside the liver or from blocked bile ducts.

Once conjugated bilirubin reaches the gut, bacteria turn it into urobilinogen and then stercobilin, which gives stool its brown colour. A little urobilinogen is absorbed and passes out in the urine. Two bedside signs follow from this. If no bilirubin reaches the gut, the stools become pale. And because conjugated bilirubin dissolves in water, it

passes into the urine and makes it dark. Unconjugated bilirubin, bound to albumin, never appears in urine. So pale stools and dark urine mean a conjugated problem.

Now put the five steps together. A rise in unconjugated bilirubin comes from production, transport or conjugation. A rise in conjugated bilirubin comes from uptake or excretion. And here is the rule that makes the whole chapter useful: a raised conjugated bilirubin is the only bilirubin result that points to disease of the liver or bile ducts. Unconjugated hyperbilirubinaemia is a problem of blood, of enzyme, or of the newborn period. Conjugated hyperbilirubinaemia is a problem of the liver.

Part two. Why newborns are yellow.

Almost every newborn becomes a little yellow in the first week of life. To understand why, compare how much bilirubin a baby makes with how much the liver can process.

A term newborn produces about six to eight milligrams of bilirubin per kilogram of body weight each day. An adult produces about four. So the newborn makes roughly twice the adult amount for its size. There are three reasons. The newborn has more red cells per kilogram. Those red cells live a shorter time, around seventy to ninety days instead of one hundred and twenty. And some conjugated bilirubin in the newborn gut is split back into unconjugated bilirubin and absorbed again.

At the same time, the enzyme UGT one A one is barely switched on. In late fetal life, its activity is only between one tenth of one percent and one percent of the adult level. After birth it climbs through several stages, and it reaches adult levels only during infancy.

So physiological jaundice is not a disease. It is a mismatch in timing: a double load arriving at a liver whose enzyme has not yet caught up. And because the bottleneck is conjugation, physiological jaundice is always unconjugated. This gives us the most important rule in neonatal hepatology. A raised conjugated bilirubin in a newborn is never physiological. It always needs an explanation.

Part three. What the laboratory actually measures.

Guidelines and clinicians use the words direct and conjugated as if they meant the same thing. They do not, and the difference matters.

The classic method is the diazo reaction, first described more than a century ago. Bilirubin reacts with a diazo reagent to form a coloured product. Conjugated bilirubin reacts quickly on its own, directly. Unconjugated bilirubin reacts only when an accelerator, such as caffeine or alcohol, is added. So the laboratory measures the total bilirubin with the accelerator, and the direct bilirubin without it. The indirect bilirubin is not measured at all. It is simply the total minus the direct.

The problem is that direct bilirubin is not a pure measure of conjugated bilirubin. It includes three things. It includes the true conjugated bilirubin. It includes a third form called delta bilirubin, which we will come to shortly. And when the total bilirubin is high, as it often is in newborns, some unconjugated bilirubin also reacts and is counted as direct. So direct bilirubin always overestimates conjugated bilirubin.

Some laboratories use a different method, which measures conjugated bilirubin itself, meaning only the monoglucuronide and diglucuronide. A common example is a dry-slide analyser that reports unconjugated and conjugated bilirubin separately. This gives a much lower number in healthy babies. In a large study of healthy term newborns by Davis and colleagues in 2011, the ninety-ninth centile for conjugated bilirubin was only zero point five. About three quarters of healthy newborns had a conjugated bilirubin of zero. For direct bilirubin, the ninety-ninth centile was two point one. That is four times higher, in the same babies.

A recent audit by Bassi and colleagues in 2025 compared the two methods on the same samples from infants under three months of age. The mean direct bilirubin was about thirty-two micromoles per litre, while the mean conjugated bilirubin was about nine. By the time the conjugated bilirubin reached seventeen micromoles per litre, which is the usual referral trigger, the direct bilirubin was already at least thirty-four.

The practical consequence is simple. Every threshold belongs to one method. If you apply a conjugated threshold to a direct result, you will call too many babies abnormal. If you apply a direct threshold to a conjugated result, you will miss babies who are truly abnormal. Before you interpret the number, find out which method your laboratory uses.

Now the third form, delta bilirubin. When conjugated bilirubin stays in the blood for a long time, some of it becomes permanently bound to albumin. This albumin-bound form is called delta bilirubin, or biliprotein. It reacts in the direct test, but it cannot be excreted by the liver or the kidney. It disappears only when the albumin itself is broken down, and albumin has a half-life of about three weeks. In prolonged hepatocellular or cholestatic jaundice, delta bilirubin can make up anything from about a tenth to nine tenths of the total. It is absent in Gilbert syndrome, in haemolysis and in physiological jaundice, because in those conditions there is no build-up of conjugated bilirubin in the first place. We will see why delta bilirubin matters at the bedside later in this chapter.

Part four. How much is a raised value worth?

Suppose a newborn has a slightly raised conjugated bilirubin. How likely is liver disease? The best answer comes from the same study by Davis and colleagues. They looked at more than two hundred and seventy thousand term newborns in Northern California and calculated likelihood ratios for liver and biliary disease at different

levels of conjugated and direct bilirubin. A likelihood ratio tells you how much a result changes the odds of disease. A value of one changes nothing. A value above ten is a strong signal.

For conjugated bilirubin, the likelihood ratios rose very steeply. A value of zero gave a ratio of about zero point four, which makes disease less likely. A value from zero point one to zero point five gave a ratio of about one, which changes nothing. A value from zero point five to just under two gave a ratio of about three. A value from two to just under five gave a ratio of about eighty. And a value of five or more gave a ratio of about five hundred. For direct bilirubin, the ratios were lower, because the test is less specific: about eighteen for values from two to just under five, and about two hundred and twenty for values of five or more.

Two findings from this study are worth remembering. Among babies with a conjugated bilirubin between zero point five and two, ninety-six percent turned out to have no liver or biliary disease. But two percent had a serious bacterial infection, such as a urinary tract infection or sepsis. And among babies with a conjugated bilirubin of five or more, almost half had biliary disease and most of the rest had liver disease.

So a mildly raised conjugated bilirubin in a well-looking newborn is a reason to look for infection and to repeat the test soon. It is not yet a reason to rush to imaging. A high value is a reason to act quickly.

Part five. What the guidelines call abnormal.

The most important reason to measure conjugated or direct bilirubin in an infant is to find biliary atresia early, because the operation for biliary atresia, the Kasai portoenterostomy, works better the earlier it is done. Five major documents give advice, and they do not agree. They differ on which fraction to measure, what number is abnormal, and at what age to test.

The joint North American and European guideline on cholestatic jaundice in infants, from NASPGHAN and ESPGHAN in 2017, uses direct bilirubin. It calls a direct bilirubin above one milligram per decilitre, or seventeen micromoles per litre, abnormal. It says that any infant who is still jaundiced after two weeks of age should have it measured.

The American Academy of Pediatrics guideline on neonatal hyperbilirubinaemia, from 2022, recommends measuring total bilirubin together with direct or conjugated bilirubin. It calls a direct bilirubin above one abnormal, or a conjugated bilirubin above zero point three. It also sets different ages for testing: at two weeks in formula-fed babies, but at three to four weeks in breastfed babies, because breast milk jaundice commonly lasts longer.

The British NICE guideline on jaundice in babies under twenty-eight days measures conjugated bilirubin. It calls a conjugated bilirubin above twenty-five micromoles per

litre, about one and a half milligrams per decilitre, a sign of possible liver disease. It applies to prolonged jaundice, meaning jaundice lasting more than fourteen days in a term baby or more than twenty-one days in a preterm baby.

The Italian society, SIGENP, in its 2021 guideline, also uses conjugated bilirubin. It calls a value above one abnormal. But when the total bilirubin is above five, it also accepts a conjugated fraction of more than twenty percent of the total as abnormal. It applies to any infant under one year of age with jaundice.

Finally, the 2025 guidance on biliary atresia screening from FISPGHAN, the federation of the world's paediatric gastroenterology societies, accepts either direct or conjugated bilirubin above one. It goes further than the others by recommending universal newborn screening, either with a stool colour card given to every family or with a bilirubin measurement in every newborn.

What do they agree on? That a raised conjugated or direct bilirubin in an infant must not be ignored, and that the infant should be referred promptly to a paediatric hepatologist or another expert. Where do they differ? The UK threshold is higher than the others. The American guideline allows breastfed babies an extra week or two before testing. And only the international federation calls for screening every newborn.

Part six. Why the twenty percent rule fails.

Many textbooks and hospital order sets still teach that cholestasis means a direct bilirubin greater than twenty percent of the total. This rule is the single most common reason why a baby with biliary atresia is sent home with reassurance.

The rule was designed for an era when newborns often had very high totals from physiological jaundice. The logic was that a direct bilirubin of two on a total of fifteen was just a small spill-over. But this logic fails in exactly the babies who matter most.

The key evidence comes from Harpavat and colleagues in 2011. They looked back at the newborn blood tests of babies who were later diagnosed with biliary atresia. Almost all of them already had a raised direct bilirubin in the first days of life. But in seventy-nine percent, the direct bilirubin was twenty percent of the total or less. In other words, by the twenty percent rule, four out of five babies with biliary atresia looked normal.

There are two reasons. First, in a newborn, a large unconjugated bilirubin from physiological jaundice makes the total big, and a big total hides a direct bilirubin that is truly abnormal. Second, the total changes with phototherapy and with feeding, so the ratio moves even when the liver has not changed.

The modern guidelines replaced the ratio with an absolute number: a direct bilirubin above one, or a conjugated bilirubin above about zero point three to zero point five,

whatever the total. Only the Italian guideline keeps the twenty percent rule, and only as a second rule when the total is above five.

At the bedside, the rule is simple: never divide. Read the direct or conjugated value itself. If the direct bilirubin is raised and the ratio is low, the ratio is wrong. For example, a direct bilirubin of one point four on a total of twelve is a ratio of about twelve percent. By the old rule, that is normal. In reality, it is biliary atresia until proved otherwise. If the value is only just above the limit and the baby is well, repeating it within a few days to two weeks improves the predictive value. But the baby should not be lost to follow-up.

Part seven. Unconjugated hyperbilirubinaemia: one gene, several diseases.

We now turn to inherited disorders, starting with those that raise unconjugated bilirubin. Gilbert syndrome, Crigler-Najjar syndrome type two and Crigler-Najjar syndrome type one are often taught as three separate diseases. It is more useful to think of them as one continuum. They are all caused by variants in the same gene, UGT one A one, and they differ in how much enzyme activity is left.

Gilbert syndrome is at the mild end. About thirty percent of normal enzyme activity remains. The usual cause is a change in the promoter, the switch region in front of the gene. The normal promoter contains six repeats of the two letters T and A. In Gilbert syndrome there are seven, and this variant is called UGT one A one star twenty-eight. People with two copies make less enzyme. In East Asia, a different variant, called star six, is more common. The bilirubin is typically around three milligrams per decilitre, and liver enzymes are normal. Gilbert syndrome carries no risk to the liver or the brain. Its only practical importance is in how some drugs are handled, which we will come back to. The right management is to reassure the family and to stop testing.

Crigler-Najjar syndrome type two is in the middle. Enzyme activity is reduced but can still be increased by drugs. It is usually caused by a promoter variant combined with a change in the coding part of the gene. The bilirubin is typically around thirteen milligrams per decilitre. There is also an intermediate group, with values around five. The classic diagnostic test is phenobarbital, which switches on the remaining enzyme. In type two, phenobarbital lowers the bilirubin by more than thirty percent. The danger in type two is kernicterus, meaning brain damage from bilirubin, during an intercurrent illness, when the bilirubin can suddenly rise. Treatment is phenobarbital, avoiding fasting, and avoiding drugs that push bilirubin off albumin.

Crigler-Najjar syndrome type one is at the severe end. There is no enzyme activity at all, because both copies of the gene are broken. Bilirubin rises by about four milligrams per decilitre every day from birth. Phenobarbital has no effect. Without treatment, kernicterus develops in the first weeks of life. These babies need intensive phototherapy from the first day, and eventually a liver transplant. Gene therapy is now being tested.

Gilbert syndrome deserves a closer look, because it is so common. In a study from Amsterdam published by Bosma and colleagues in 1995, about forty percent of the alleles in healthy people carried the seven-repeat promoter. Across different ancestries, promoters with anything from five to eight repeats are found, and the more repeats, the lower the activity.

On its own, the Gilbert variant is harmless. But it can become important when a second problem is added. The best example is G six P D deficiency, a common inherited cause of haemolysis. In a study by Kaplan and colleagues in 1997, neither G six P D deficiency alone nor the Gilbert variant alone increased the rate of significant jaundice in newborns. But together they did. Among babies with G six P D deficiency who carried one copy of the Gilbert variant, about thirty-two percent developed significant jaundice. With the normal promoter, the figure was about ten percent. And with two copies of the Gilbert variant, it was fifty percent. The lesson is that more production plus slower conjugation is much worse than either alone.

One practical warning. The standard genetic test for Gilbert syndrome counts the TA repeats. In East Asian children, the usual cause is the star six variant, which this test will miss.

Part eight. Crigler-Najjar type one: what happens without treatment, and what is new.

The best natural history of Crigler-Najjar type one comes from a study by Strauss and colleagues in 2020, at the Clinic for Special Children in Pennsylvania. This clinic serves Amish and Mennonite communities, where a single founder variant is common. They described twenty-eight patients, with more than five hundred patient-years of follow-up between them.

In newborns who had not yet received phototherapy, the total bilirubin rose by about four milligrams per decilitre each day. Four of the twenty-eight, or fourteen percent, developed kernicterus, between two and six weeks of age. Brain injury was predicted by a bilirubin-to-albumin molar ratio of one or more, or by a total bilirubin of thirty or more. Babies whose phototherapy started after the thirteenth day of life had three and a half times the risk of brain injury.

Keeping these children safe required a great deal of light: on average about nine hours of phototherapy every day, for years. Even so, bilirubin crept up by about half a milligram per decilitre each year. The reason is that as a child grows, the body gets bigger but the skin surface available to the light does not keep pace. So adolescence, not infancy, is when phototherapy begins to fail. Seventeen of the twenty-eight patients received a liver transplant, at a median age of about sixteen years.

In 2023, D'Antiga and colleagues reported the first gene therapy trial for Crigler-Najjar syndrome in the New England Journal of Medicine. Five patients received a

single infusion of a virus vector, AAV eight, carrying a working copy of UGT one A one. The lower dose failed within four months. The higher dose worked. In that group, the mean bilirubin fell from about three hundred and fifty micromoles per litre to about one hundred and fifty, and the patients stayed off phototherapy for at least a year and a half. Four of the five had a rise in ALT, which was treated with steroids, and there were no serious adverse events.

There is an important limitation. This kind of vector does not become part of the cell's own DNA. When the liver grows, the vector is diluted among new cells. So we do not yet know whether gene therapy given in infancy would last. For now, liver transplantation remains the cure, and gene therapy is not yet an alternative for a newborn.

Part nine. Isolated conjugated hyperbilirubinaemia: Dubin-Johnson and Rotor.

Two rare inherited disorders cause a raised conjugated bilirubin with completely normal liver enzymes. Both are benign and do not progress. Their main importance is that recognising them avoids unnecessary and invasive tests.

Dubin-Johnson syndrome is caused by variants in the gene *ABCC2*, which codes for the pump MRP2. Conjugated bilirubin is made normally, but it cannot leave the liver cell into bile. A dark pigment builds up in the liver cells, and at surgery or biopsy the liver looks black.

Rotor syndrome is caused by loss of both transporters, *OATP1B1* and *OATP1B3*. Conjugated bilirubin that is pumped back into the blood cannot be taken up again. The liver looks normal and contains no pigment.

The test that separates them is the measurement of coproporphyrins in the urine. Coproporphyrins are by-products of haem synthesis, and there are two forms, called isomer one and isomer three. In Dubin-Johnson syndrome, the total amount in the urine is normal, but more than eighty percent is isomer one. In Rotor syndrome, the total is raised two to five fold, and about sixty-five percent is isomer one.

Both conditions have a normal outlook. Recognising Dubin-Johnson syndrome can save a child from a liver biopsy and a cholangiogram. Recognising Rotor syndrome also has a pharmacological meaning: the same transporters carry many drugs, including statins and methotrexate, so their handling is altered.

Dubin-Johnson syndrome can present in the newborn period, where it looks like any other cholestasis. A study from Riyadh by Al-Hussaini and colleagues in 2023 found a disease-causing *ABCC2* variant in twenty-eight of five hundred and thirty-three infants with cholestasis. Ninety-two percent of them had a normal ALT. In this setting, a normal ALT had a sensitivity of ninety-three percent for Dubin-Johnson

syndrome, and a negative predictive value of more than ninety-nine percent. A urinary coproporphyrin isomer one fraction above eighty percent was one hundred percent sensitive and specific. The median fraction was eighty-eight percent in Dubin-Johnson syndrome and sixty-seven percent in other causes of neonatal cholestasis.

A study from Hannover by Junge and colleagues in 2021 compared infants with Dubin-Johnson syndrome and infants with biliary atresia. The infants with Dubin-Johnson syndrome had lower AST, ALT and GGT. But stool colour, bile acids and total bilirubin did not separate the two groups. So a pale-looking stool does not rule out Dubin-Johnson syndrome, and it does not rule in biliary atresia.

The practical message is this. In an infant with cholestasis whose liver enzymes and GGT are normal, send urinary coproporphyrins before a liver biopsy. Remember, though, that a normal GGT in cholestasis has other important causes too, including progressive familial intrahepatic cholestasis and bile acid synthesis disorders, and these are not benign.

Part ten. Reading bilirubin like an expert.

Four observations separate an expert reading from a routine one.

The first is delta bilirubin. After a hepatitis or an episode of cholestasis, the jaundice often lasts for weeks after the liver has recovered. The reason is delta bilirubin, stuck to albumin, which disappears only as fast as albumin is broken down, with a half-life of about three weeks. So the child looks more yellow than the liver deserves. A useful clue is that the urine is no longer dark, because delta bilirubin cannot pass into the urine. If the child is well, the enzymes are falling and the INR is normal, a slowly falling bilirubin is expected and does not mean the disease is continuing.

The second is the haemolysed sample. Haemolysis in the test tube interferes with the measurement and lowers the direct bilirubin. So a haemolysed specimen can hide cholestasis. If the sample is haemolysed, take a new one rather than reassure the family.

The third is fasting and illness. In Gilbert syndrome, the bilirubin can double with fasting, fever, heavy exercise or an intercurrent illness. The same is true of the bilirubin measured in a child after surgery or in intensive care. Before you label a child with anything, repeat the test when the child is well and has eaten normally.

The fourth is pharmacogenetics. People with two copies of the Gilbert variant, star twenty-eight, handle some drugs less well. The best-known example is irinotecan, a chemotherapy drug, where the starting dose should be reduced. Atazanavir, an HIV drug, inhibits UGT one A one and causes visible jaundice in the same people. And variants in the Rotor transporters change the levels of statins and methotrexate.

To summarise the approach in one sentence: read the fraction first, then the size of the number, then the trend over time, and always read the albumin beside it.

Part eleven. Four clinical cases.

Case one. A three-week-old breastfed boy is still jaundiced. He is feeding well and gaining weight. His total bilirubin is twelve and his direct bilirubin is one point four. The registrar says the direct fraction is only twelve percent of the total, so this is breast milk jaundice. Is that right? No. The ratio does not matter. A direct bilirubin of one point four is above the limit of one, and in a three-week-old it is biliary atresia until proved otherwise. Look at the stools yourself, check the urine, and refer the same week to a paediatric hepatologist for an ultrasound and further work-up.

Case two. A fifteen-year-old boy notices yellow eyes after three days of gastroenteritis, when he was hardly eating. His total bilirubin is three point one, and his direct bilirubin is zero point three. His ALT, alkaline phosphatase, GGT, blood count and reticulocyte count are all normal. What is the most likely diagnosis? Gilbert syndrome. The raised bilirubin is unconjugated, there is no sign of haemolysis or liver disease, and fasting makes Gilbert syndrome worse. Repeat the test when he is well to confirm the pattern. Then reassure him and his family, and stop testing. Genetic testing is not needed, but it is worth recording the diagnosis in case he later needs drugs such as irinotecan.

Case three. A six-week-old girl has conjugated hyperbilirubinaemia, with a direct bilirubin of three. She looks well. Her ALT, AST and GGT are all normal. Her stools are pale yellow. What test should you consider before a liver biopsy? Urinary coproporphyrins. A cholestatic infant with normal enzymes and normal GGT may have Dubin-Johnson syndrome, and an isomer one fraction above eighty percent would confirm it. But normal GGT cholestasis also includes serious diseases, such as progressive familial intrahepatic cholestasis and bile acid synthesis defects, so she needs full assessment by a hepatologist. And pale stools always need biliary atresia to be excluded.

Case four. A newborn girl from a consanguineous family becomes deeply jaundiced on the second day of life. Her total bilirubin rises by about four each day. Her direct bilirubin is low, and there is no haemolysis. What should you suspect, and what matters most now? Crigler-Najjar syndrome type one. What matters most is to start intensive phototherapy at once, to follow the bilirubin-to-albumin ratio, and to avoid drugs that displace bilirubin. Starting phototherapy late greatly increases the risk of kernicterus. Genetic testing of UGT one A one confirms the diagnosis, and the family should be introduced to a transplant centre early.

Key points to remember.

A raised conjugated bilirubin is the only bilirubin result that points to liver or biliary disease. A raised unconjugated bilirubin points to blood, enzyme or the newborn period.

Bilirubin handling has five steps: production, transport on albumin, uptake, conjugation by UGT one A one, and excretion by MRP two. Each has its own diseases.

Physiological jaundice is always unconjugated. A raised conjugated bilirubin in a newborn is never physiological.

Direct bilirubin and conjugated bilirubin are different tests. Direct always reads higher. Use the threshold that belongs to your laboratory's method.

Most guidelines call a direct or conjugated bilirubin above one milligram per decilitre abnormal in an infant. The UK uses a conjugated bilirubin above twenty-five micromoles per litre.

Never use the twenty percent rule. Four out of five newborns with biliary atresia have a ratio of twenty percent or less.

A mildly raised conjugated bilirubin in a well newborn should make you look for infection and repeat the test.

Gilbert syndrome and Crigler-Najjar syndrome types one and two are one spectrum of UGT one A one activity. Phenobarbital lowers the bilirubin in type two, but not in type one.

In Crigler-Najjar type one, start phototherapy from day one. Phototherapy fails in adolescence, and transplantation is still the cure.

Dubin-Johnson and Rotor syndromes cause benign conjugated jaundice with normal enzymes. Urinary coproporphyrins separate them and can prevent a biopsy.

Jaundice can outlast the disease because of delta bilirubin. A haemolysed sample can hide cholestasis. Fasting and illness raise a Gilbert bilirubin.

Questions to test yourself.

Question one. Which bilirubin fraction points to disease of the liver or bile ducts, and why? Answer: the conjugated fraction. A rise means that bilirubin has been processed by the liver but cannot leave through the bile, or cannot be taken back up.

Question two. Why do newborns become jaundiced in the first week? Answer: they produce about twice as much bilirubin per kilogram as adults, and their UGT one A one enzyme is still immature. The result is unconjugated jaundice.

Question three. A laboratory changes from a direct bilirubin method to a conjugated bilirubin method, but keeps the old cut-off of one milligram per decilitre. What will happen? Answer: some truly abnormal babies will be missed, because conjugated bilirubin reads much lower than direct. The cut-off must match the method.

Question four. A four-week-old has a total bilirubin of ten and a direct bilirubin of one point six. Is the twenty percent rule reassuring? Answer: no. The ratio is sixteen percent, but the direct value itself is abnormal. This infant needs urgent evaluation for biliary atresia.

Question five. How can you tell Crigler-Najjar type two from type one at the bedside? Answer: give phenobarbital. In type two the bilirubin falls by more than thirty percent. In type one it does not change.

Question six. A child is recovering well from hepatitis A. The ALT is almost normal and the urine is clear, but she is still yellow three weeks later. Why? Answer: delta bilirubin, which is bound to albumin and disappears only with albumin's half-life of about three weeks.

Question seven. What test separates Dubin-Johnson syndrome from Rotor syndrome? Answer: urinary coproporphyrins. In Dubin-Johnson syndrome the total is normal and more than eighty percent is isomer one. In Rotor syndrome the total is raised and about sixty-five percent is isomer one.

End of chapter.